MAGNETIC STORAGE: What Lies Ahead

The future promises hard disk drives with increased bit density and smaller form factors

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If there's one system component that has seen dramatic and continuous change over the years, it's the storage subsystem. Ever since the debut of the first clunky 5 MB hard disk drives, the increase in the capacity and speed of these drives has been spiralling upwards. With the latest breed of hard disks incorporating cutting edge implementations in the fields of mechanics, electronics, chemical engineering, materials and even aerodynamics, hard disk drives are a good indicator of the state of today's technology.

Most people don't really give too much thought to a hard disk in the overall reckoning of which computer to buy—they generally go in for one that fits their budget without bothering with details such as the interface or the drive's spindle speed. However, these factors can make a world of difference, especially while working with applications such as image processing, sound and video editing and hardcore gaming.

This is becoming more pronounced with today's applications being more demanding on your computer's resources than was the case in the past. Everything from the operating system's boot-up time to the startup time taken by an application and virtual memory performance is directly affected by your hard disk's speed. Today's drives are considerably faster. With the newer 7,200-rpm hard disk drives, the overall system can operate in a smoother and faster fashion.

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Another very pertinent issue is that of noise levels. At home or work, noise is a very important factor to consider—the continuous whirr of a hard disk drive can really spoil your movie-watching experience or disturb your concentration when you are trying to type those reports! Today's hard disks incorporate a range of enhancements in their construction and in the implementation of materials used, driving their noise level down to just above the threshold of human hearing.

Companies have also realised that as far as the average hard disk user is concerned, it's not just about capacity any more—most users do not even use the entire hard disk during the course of their computer's life. It's now all about how to get the maximum performance into the same price bracket that consumers are used to paying.

On the technology front, there have been a range of improvements along with new standards that have contributed to the superior performance and reliability of hard disk drives today. The most important among these is the increase in the 'areal density' or the number of bits that can be packed onto the surface of a hard disk's platter. The current standard is 40 GB per platter, which will increase to 60 GB per platter by the end of this year and further on to a whopping 80 GB per platter by the beginning of the next year.

This is going to allow much larger capacity hard disks to be fabricated using lesser number of platters and this consequently translates into lower prices for the end-user. With increasing bit density, there is also going to be a simultaneous improvement in the technology used in the drive's read-write heads (from 0.18 micron to 0.13 micron). Also on the board are new interfaces such as Serial ATA and USB 2.0—interfaces that promise performance while still being simple to configure and use.

The most important factor that indicates the direction of the hard disk industry is the application it's used for. All this while hard disks were used only in desktops or servers, but with the proliferation of 'lifestyle' products, you will now see an increasing number of hard disks used in non-PC applications—in devices such as digital cameras, MP3 players, set-top boxes, digital video recorders and gaming consoles.

Soon hard disk drives with smaller form factors that can be used in these applications will appear in increasing numbers—in the coming year, about 20 per cent of hard disks sold in the desktop category will be used for non-PC applications. This is indicative of a very significant trend in the use and the direction of magnetic storage in the years to come.

As told to Marco D'Souza